

Build a Document Question Answering (Document QA) Agent

Develop an AI-powered Document QA agent that enables users to upload a document from their local system and ask questions based on its content. The agent should extract the document content, understand the user's query, and provide accurate answers using only the uploaded document as the source of information. If the requested information is not available in the document, the agent should respond appropriately instead of generating unsupported answers.

Expected Outcome: Users should be able to upload a document, ask multiple questions, and receive context-aware answers derived only from the uploaded document.

list of nodes :

- 1.Manual Trigger
- 2.Read /write files from disk
- 3.Extract from file
- 4.Basic LLM chain

The screenshot displays the n8n workflow editor interface. The workflow is titled "Document Extractor" and is located in the "Personal" workspace. It consists of four nodes connected in a sequence:

- When clicking 'Execute workflow'**: A manual trigger node.
- Read/Write Files from Disk**: A node to read files from the local disk.
- Extract from File**: A node to extract content from a PDF file.
- Basic LLM Chain**: A node to process the extracted content using a Large Language Model (LLM).

The workflow is currently in the "Editor" tab. Below the workflow, the "Logs" section shows the execution details for the "Basic LLM Chain" node, which was successful in 1.125s and used 780 Tokens. The "INPUT" section shows the following data:

```
# numpages 2
# numrender 2
info
  PDFFormatVersion 1.5
  Language en-US
  EncryptFilterName [null]
  IsLinearized false
  IsAcroFormPresent false
```

The "OUTPUT" section shows the following text:

```
text
The error should now be resolved. **Check Against Constraints:** - Uses only provided document? Yes. - Answers the specific query? Yes. - No external info? Yes. - Format/ tone appropriate? Yes. All constraints met. Ready to output. All steps verified. Output matches draft. Based on the provided document, the 4th step of the solution is: **4. Verify the Fix** - Reopen your terminal or code editor (e.g., VS Code) and try running your npm command again. The error should now be resolved.
```

The screenshot displays the n8n workflow editor for a workflow named 'Document Extractor - n8n'. The workflow is composed of four nodes connected sequentially:

- When clicking 'Execute workflow'**: The starting trigger node.
- Read/Write Files from Disk**: A node to read files from the disk.
- Extract from File**: A node to extract content from a PDF file.
- Basic LLM Chain**: A node that processes the extracted content using a Groq Chat Model.

The 'Basic LLM Chain' node has been executed successfully, as indicated by the 'Workflow executed successfully' message. The output of the node is displayed in the 'OUTPUT' section:

```

text
<think> Here's a thinking process: 1. **Analyze User Input:** - **Document Content:** Provides troubleshooting steps for a '.ps1' error related to PowerShell Execution Policy. It lists 4 steps: 1. Open PowerShell as Administrator, 2. Check Current Policy, 3. Change Execution Policy, 4. Verify the Fix. - **Instruction:** Extract content, understand query, answer accurately using *only* the document. If info isn't there, respond appropriately. DO NOT take answers outside the document. - **User's Query:** "WHAT IS THE 4th step of SOLUTION FOR Troubleshooting .ps1 Error?" 2. **Locate Relevant Information in Document:** - I need to find the 4th step in the "Solution Steps" section. - The document clearly
  
```

The 'INPUT' section for the 'Basic LLM Chain' node shows the following data:

```

# numpages 2
# numrender 2
info
  T PDFFormatVersion 1.5
  T Language en-US
  EncryptFilterName [null]
  IsLinearized false
  IsAcroFormPresent false
  IsXFAPresent false
  IsCollectionPresent false
  IsSignaturesPresent false
  T Author Harrish Alaguponniah
  T Creator Microsoft® Word 2016
  T CreationDate D:20260717150802+05'30'
  
```

The screenshot shows the configuration of the 'Basic LLM Chain' node in the n8n workflow editor. The 'Parameters' tab is selected, displaying the following settings:

- Save time with an example of how this node works**: A checkbox that is checked.
- Source for Prompt (User Message)**: A dropdown menu set to 'Define below'.
- Prompt (User Message)**: A text area containing the following prompt:


```

Document content : {{ $json.text }}
Instruction : The agent should extract the document content, understand the user's query, and provide accurate answers using only the uploaded document as the source of information. If the requested information is not available in the document, the agent should respond appropriately instead of generating unsupported answers.
Document content : Troubleshooting .ps1 Error It means...
      
```
- Require Specific Output Format**: A toggle switch that is turned off.
- Enable Fallback Model**: A toggle switch that is turned off.
- Chat Messages (if Using a Chat Model)**: A text area that is currently empty.
- Model ***: A dropdown menu set to 'Groq Chat Model'.

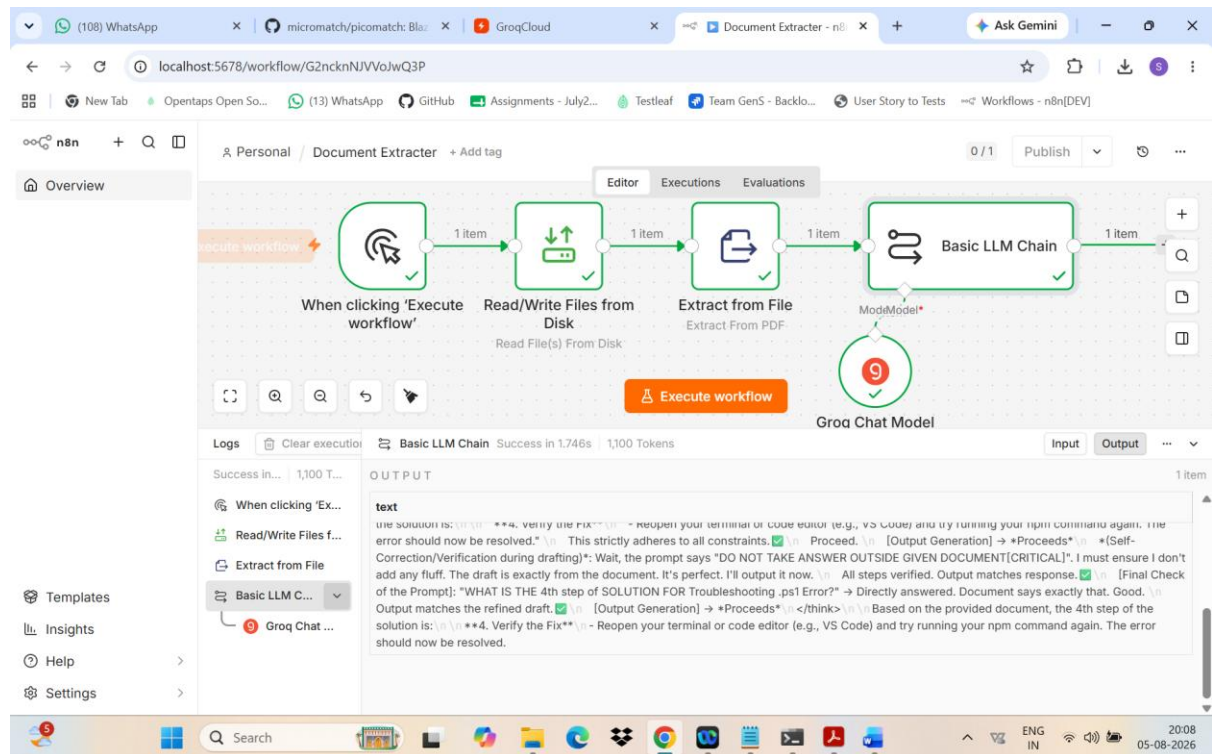
The 'OUTPUT' section on the right shows the result of the execution, which is a detailed analysis of the PowerShell error, matching the output shown in the first screenshot.

Prompt : Document content : {{ \$json.text }}

Instruction : The agent should extract the document content, understand the user's query, and provide accurate answers using only the uploaded document as the source of information. If the requested information is not available in the document, the agent should respond appropriately instead of generating unsupported answers.

Expected Outcome: User will be asked Query/question ask multiple questions, and receive context-aware answers derived only from the uploaded document.D0 NOT TAKE ANSWER OUTSIDE GIVEN DOCUMENT[CRITICAL]

user's Query : WHAT IS THE 4th step of SOLUTION FOR Troubleshooting .ps1 Error?



Output:

<think>\nHere's a thinking process:\n\n1. **Analyze User Input:**\n - **Document Content:** Provides troubleshooting steps for a '.ps1' error related to PowerShell Execution Policy. It lists 4 steps: 1. Open PowerShell as Administrator, 2. Check Current Policy, 3. Change Execution Policy, 4. Verify the Fix.\n - **Instruction:** Extract content, understand query, answer accurately using *only* the document. If info isn't there, respond appropriately. DO NOT take answers outside the document.\n - **User's Query:** "WHAT IS THE 4th step of SOLUTION FOR Troubleshooting .ps1 Error?"\n\n2. **Locate Relevant Information in Document:**\n - I need to find the 4th step in the "Solution Steps" section.\n - The document clearly lists:\n "4. Verify the Fix\n • Reopen your terminal or code editor (e.g., VS Code) and try running your npm command again. The error should now be resolved."\n\n3. **Formulate Response:**\n - Must be direct, accurate, and strictly based on the provided text.\n - State the 4th step clearly as written in the document.\n - Ensure no external information is added.\n - Match the query's focus on the "4th step of SOLUTION".\n\nDraft: Based on the provided document, the 4th step of the solution for troubleshooting the .ps1 error is:\n **4. Verify the Fix**\n - Reopen your terminal or code editor (e.g., VS Code) and try running your npm command again. The error should now be resolved.\n\n4. **Check Against Constraints:**\n -

Extracts document content? Yes.\n - Understands query? Yes.\n - Uses only uploaded document? Yes.\n - No outside info? Yes.\n - Answers the specific question? Yes.\n\n The response is ready. It's concise and directly answers the question using only the provided text. I will format it clearly. ✓\n Self-Correction/Refinement during thought: I should make sure the formatting matches the document's structure but keeps it clean. The draft looks good. I'll output it. ✓\n Final Output Generation. \n "Based on the provided document, the 4th step of the solution is:\n\n **4. Verify the Fix**\n - Reopen your terminal or code editor (e.g., VS Code) and try running your npm command again. The error should now be resolved." \n This strictly adheres to all constraints. ✓\n Proceed. \n [Output Generation] -> *Proceeds*\n *(Self-Correction/Verification during drafting)*: Wait, the prompt says "DO NOT TAKE ANSWER OUTSIDE GIVEN DOCUMENT[CRITICAL]". I must ensure I don't add any fluff. The draft is exactly from the document. It's perfect. I'll output it now. \n All steps verified. Output matches response. ✓\n [Final Check of the Prompt]: "WHAT IS THE 4th step of SOLUTION FOR Troubleshooting .ps1 Error?" -> Directly answered. Document says exactly that. Good. \n Output matches the refined draft. ✓\n [Output Generation] -> *Proceeds*\n</think>\n\nBased on the provided document, the 4th step of the solution is:\n\n**4. Verify the Fix**\n- Reopen your terminal or code editor (e.g., VS Code) and try running your npm command again. The error should now be resolved.

Negative Testing

<think> Here's a thinking process:

1. Analyze User Input:

- **Document Content:** A troubleshooting guide for a .ps1 error related to PowerShell Execution Policy. It provides steps to open PowerShell as admin, check policy, change it to RemoteSigned, and verify the fix.
- **Instruction:** The agent must extract document content, understand the query, and provide accurate answers *using only the uploaded document*. If info isn't in the document, respond appropriately instead of generating unsupported answers. DO NOT TAKE ANSWER OUTSIDE GIVEN DOCUMENT [CRITICAL].
- **User's Query:** "who is the president of USA?"

2. Evaluate Query against Document:

- Does the document mention the President of the USA? No.
- The document is strictly about troubleshooting PowerShell execution policy errors.

3. Apply Constraints:

- Constraint: "If the requested information is not available in the document, the agent should respond appropriately instead of generating unsupported answers."
- Constraint: "DO NOT TAKE ANSWER OUTSIDE GIVEN DOCUMENT[CRITICAL]"

4. **Formulate Response:**

- Acknowledge the query.
- State clearly that the requested information is not available in the provided document.
- Reiterate that the document only covers troubleshooting PowerShell .ps1 execution policy errors.
- Maintain a helpful but strict adherence to the constraint.

Draft: Based on the provided document, there is no information about the President of the USA. The document exclusively covers troubleshooting steps for PowerShell .ps1 execution policy errors. If you have questions related to the PowerShell troubleshooting guide, I'd be happy to help with those!

5. **Check against Constraints:**

- Uses only document content? Yes, explicitly states the document doesn't contain it.
- Responds appropriately when info is missing? Yes.
- Does not generate outside info? Yes.
- Matches expected outcome? Yes.

The response is solid and strictly adheres to the prompt's critical instructions. I will output this. 
</think>

Based on the provided document, there is no information regarding the President of the USA. The document exclusively covers troubleshooting steps for PowerShell .ps1 execution policy errors.

If you have any questions related to the PowerShell troubleshooting guide or need help with the steps outlined in the document, feel free to ask!